

SOLUTIONS: TRIGONOMETRIC INTEGRATION: ADVANCED PROBLEM SHEET

Hints and solutions for the problem sheet 'trigonometric integration: advanced problem sheet'. Marking is available upon request.

Hints. Available for most but not necessarily all questions. Worked solutions can be found overleaf.

1a. Recall $\sin(2x) = 2 \sin(x) \cos(x)$.

1c. Extra constants can be absorbed into the arbitrary constant C .

2a, 2b. The most important identity here is $\cos(2x) = \cos^2(x) - \sin^2(x)$. This has two additional forms that can be obtained by using $\sin^2(x) + \cos^2(x) = 1$ - that is, use $\cos^2(x) = 1 - \sin^2(x)$ and $\sin^2(x) = 1 - \cos^2(x)$.

2c. Use the identity involving $1, \tan^2(x)$, and $\sec^2(x)$.

3a. Let $u = \operatorname{cosec}(x)$.

3b. Integration by parts, let $u = x$ and $dv/dx = \cos(3x)$.

4a. Integration by parts again, but a tricky one. Define the original integral as some variable, e.g. I , and look out for it appearing after the second integration by parts.

4b. Write $\tan(x) = \sin(x)/\cos(x)$.

4c. Let $u = \tan(x)$.

5a. Factorise $\sin^3(x) = \sin(x) \cdot \sin^2(x)$ and then use the identity $\sin^2(x) = 1 - \cos^2(x)$.

5b. Integration by parts.

5c. Write $\sin^2(x) \cos^2(x) = (\sin(x) \cos(x))^2$ and then use the double angle identity $\sin(2x) = 2 \sin(x) \cos(x)$.

Solutions.

1. The problem concerns the integral

$$\int 2 \sin(x) \cos(x) dx.$$

To (a) solve by double angle, use $\sin(2x) = 2 \sin(x) \cos(x)$ so the integral becomes a simple chain-rule case,

$$\int 2 \sin(x) \cos(x) dx = \int \sin(2x) dx = -\frac{1}{2} \cos(2x) + C.$$

To (b) solve with the substitution $u = \sin(x)$, recognise $du/dx = \cos(x)$, so the integral becomes

$$\int 2u du = u^2 + C = \sin^2(x) + C.$$

(c) To see that the answers are equivalent, use the double angle formula one more time: $\cos(2x) = \cos^2(x) - \sin^2(x)$. This has two alternative forms, $2 \cos^2(x) - 1$ and $1 - 2 \sin^2(x)$ (using the identity $\sin^2(x) + \cos^2(x) = 1$). The useful one here is the $\cos(2x) = 1 - 2 \sin^2(x)$, since then

$$-\frac{1}{2} \cos(2x) = -\frac{1}{2} (1 - 2 \sin^2(x)) = -\frac{1}{2} + \sin^2(x) = \sin^2(x) + C,$$

i.e. the constant C absorbs the constant $-1/2$. So the answers are indeed equivalent.

2. Integrating $\sin^2(x)$ and $\cos^2(x)$ is a common problem in exams. They are quite unique, both relying on the double angle formula used in Q1,

$$\begin{aligned} \cos(2x) &= \cos^2(x) - \sin^2(x) \\ &= 2 \cos^2(x) - 1 \\ &= 1 - 2 \sin^2(x). \end{aligned}$$

Rearranging, we can write

$$\begin{aligned} \sin^2(x) &= \frac{1 - \cos(2x)}{2} = \frac{1}{2} - \frac{\cos(2x)}{2}, \\ \cos^2(x) &= \frac{\cos(2x) + 1}{2} = \frac{\cos(2x)}{2} + \frac{1}{2}. \end{aligned}$$

Substituting these new identities into the integrals make them quite simple,

$$\int \sin^2(x) dx = \int \left(\frac{1}{2} - \frac{\cos(2x)}{2} \right) dx = \frac{1}{2}x - \frac{1}{4} \sin(2x) + C$$

and

$$\int \cos^2(x) dx = \int \left(\frac{\cos(2x)}{2} + \frac{1}{2} \right) dx = \frac{1}{4} \sin(2x) + \frac{1}{2}x + C.$$

The third case of $\int \tan^2(x) dx$ is completely different. Here we need to use the identity $1 + \tan^2(x) = \sec^2(x)$ to rewrite the integral

$$\int \tan^2(x) dx = \int (\sec^2(x) - 1) dx = \tan(x) - x + C.$$

3a. This integral needs the substitution $u = \operatorname{cosec}(x)$. Then $du/dx = -\operatorname{cosec}(x) \cot(x)$, and the integral becomes

$$\begin{aligned}\int \operatorname{cosec}^7(x) \cot(x) dx &= \int -\operatorname{cosec}^6(x) \cdot -\operatorname{cosec}(x) \cot(x) dx \\ &= \int -u^6 du = -\frac{1}{7}u^7 + C = -\frac{1}{7}\operatorname{cosec}^7(x) + C.\end{aligned}$$

(This is the reverse chain rule).

3b. For this integral we can apply integration by parts,

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx.$$

Using the acronym LIATE (log, inverse, algebra, trig, exponential) - or any preferred method to determine how to choose u - we set $u = x$ and $dv/dx = \cos(3x)$. Then $du/dx = 1$ and $v = 1/3 \sin(3x)$. Putting this together,

$$\begin{aligned}\int x \cos(3x) dx &= x \cdot \frac{1}{3} \sin(3x) - \int \frac{1}{3} \sin(3x) \cdot 1 dx \\ &= \frac{1}{3} x \sin(3x) + \frac{1}{9} \cos(3x) + C.\end{aligned}$$

(This is the reverse product rule).

4a. This is probably the hardest integral on the sheet. It is a special case of integration by parts where you need to 'go back on yourself'. For these types of problems it is easiest to define the integral you want to solve, i.e. define a variable I so that

$$I = \int e^x \sin(x) dx.$$

Using LIATE again, we choose $u = \sin(x)$ and $dv/dx = e^x$. Then $du/dx = \cos(x)$ and $v = e^x$. Putting this together,

$$\begin{aligned}I &= \int e^x \sin(x) dx = \sin(x)e^x - \int e^x \cos(x) dx \\ &=: e^x \sin(x) - I_2.\end{aligned}$$

To be clear, we have defined a new variable I_2 as the integral $\int e^x \cos(x) dx$. For this we need to use integration by parts again, now with $u = \cos(x)$ and $dv/dx = e^x$. So $du/dx = -\sin(x)$ and again $v = e^x$, meaning

$$\begin{aligned}I_2 &= \int e^x \cos(x) dx = \cos(x)e^x - \int e^x \cdot -\sin(x) dx \\ &= \cos(x)e^x + \int e^x \sin(x) dx \\ &= \cos(x)e^x + I,\end{aligned}$$

because the integral appearing here is the same as the original integral we called I . Without defining the original integral as I , students often feel stuck in this nested-integral loop. However, since we have defined the integral, it is clearer to see that

$$I = e^x \sin(x) - I_2 = e^x \sin(x) - (\cos(x)e^x + I).$$

Solving the equation as one normally would,

$$2I = e^x \sin(x) - e^x \cos(x) \implies I = \frac{1}{2}e^x(\sin(x) - \cos(x)),$$

and therefore

$$\int e^x \sin(x) dx = \frac{1}{2}e^x(\sin(x) - \cos(x)) + C.$$

4b. This is another special integral that needs one to write

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx.$$

This is a fraction where the numerator is (almost) the derivative of the denominator, off by a factor of -1 . Therefore

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx = -\ln(|\cos(x)|) + C.$$

4c. Another substitution! Write $u = \tan(x)$ and recognise that $du/dx = \sec^2(x)$. So then

$$\int \tan^6(x) \sec^2(x) dx = \int u^6 du = \frac{1}{7} \tan^7(x) + C.$$

5a. Most cubic (or more generally odd-powered) trig integrals need you to factor out one power and then use the celebrated identity $\sin^2(x) + \cos^2(x) = 1$. In this case,

$$\int \sin^3(x) dx = \int \sin(x) \sin^2(x) dx = \int \sin(x)(1 - \cos^2(x)) dx,$$

and then expanding the terms gives

$$\begin{aligned} \int \sin^3(x) dx &= \int \sin(x) dx - \int \sin(x) \cos^2(x) dx \\ &= -\cos(x) + \frac{1}{3} \cos^3(x) + C. \end{aligned}$$

The second integral can be seen through the substitution $u = \cos(x)$.

5b. This is another problem to be solved with integration by parts. Choose $u = x$ and $dv/dx = \sec^2(x)$. Then $du/dx = 1$ and $v = \tan(x)$, so then

$$\begin{aligned} \int x \sec^2(x) dx &= x \tan(x) - \int \tan(x) dx \\ &= x \tan(x) + \ln(|\cos(x)|) + C, \end{aligned}$$

where $\int \tan(x)dx$ is found by considering $\tan(x) = \sin(x)/\cos(x)$.

5c. The final integral of this worksheet! We again use the double angle identity $\sin(2x) = 2\sin(x)\cos(x)$. To see this, write

$$\begin{aligned}\int \sin^2(x) \cos^2(x) dx &= \int (\sin(x) \cos(x))^2 dx \\ &= \int \left(\frac{1}{2} \sin(2x)\right)^2 dx = \frac{1}{4} \int \sin^2(2x) dx.\end{aligned}$$

Then it becomes a case of integrating $\sin^2$, done earlier in Q2a. Following the same logic,

$$\int \sin^2(2x) dx = \frac{1}{2}x - \frac{1}{8} \sin(4x) + C,$$

so to finish

$$\int \sin^2(x) \cos^2(x) dx = \frac{1}{8}x - \frac{1}{32} \sin(4x) + C.$$